

EMPLOYEE RETENTION PREDICTIVE MODELS FOR HR GROUPS

USING IBM SPSS MODELER

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Introduction

Employee Retention Predictive Models for HR Groups

Our project focuses on predicting employee attrition using machine learning in IBM SPSS Modeler, helping HR identify at-risk employees and support retention strategies.

By using predictive models, HR departments can make better strategic decisions, improve employee satisfaction, and create a positive work environment. The project demonstrates how data analytics can help organizations reduce turnover rates and improve workforce management efficiently.



OBJECTIVE OF THE PROJECT

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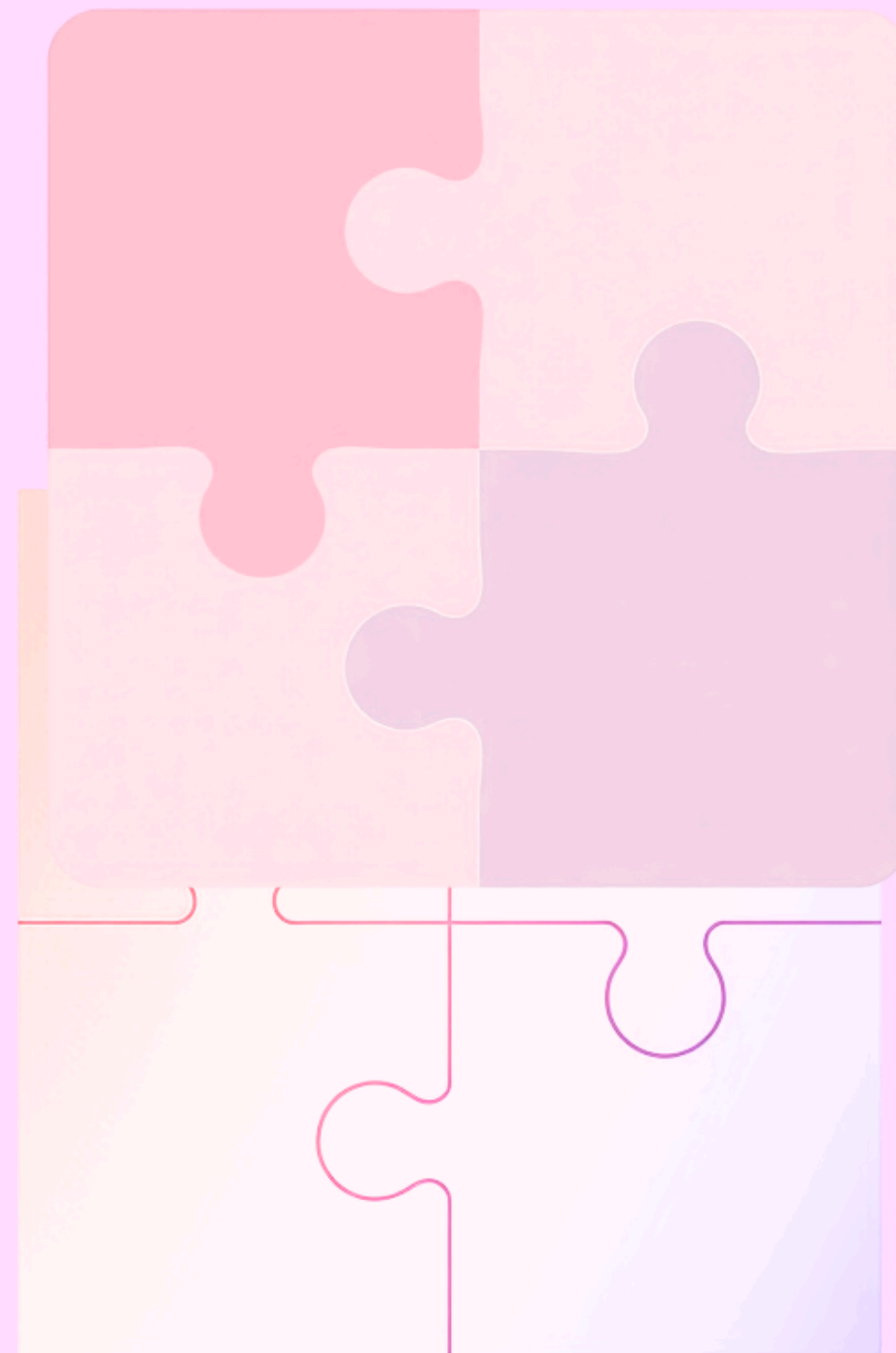
Strengths

Analyze employee data to identify key attrition factors and patterns that drive turnover within the organization.

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Weaknesses

Build predictive models for retention using machine learning to forecast which employees are at risk of leaving.



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Threats

High employee turnover impacts productivity, increases hiring costs, and reduces operational efficiency across teams.

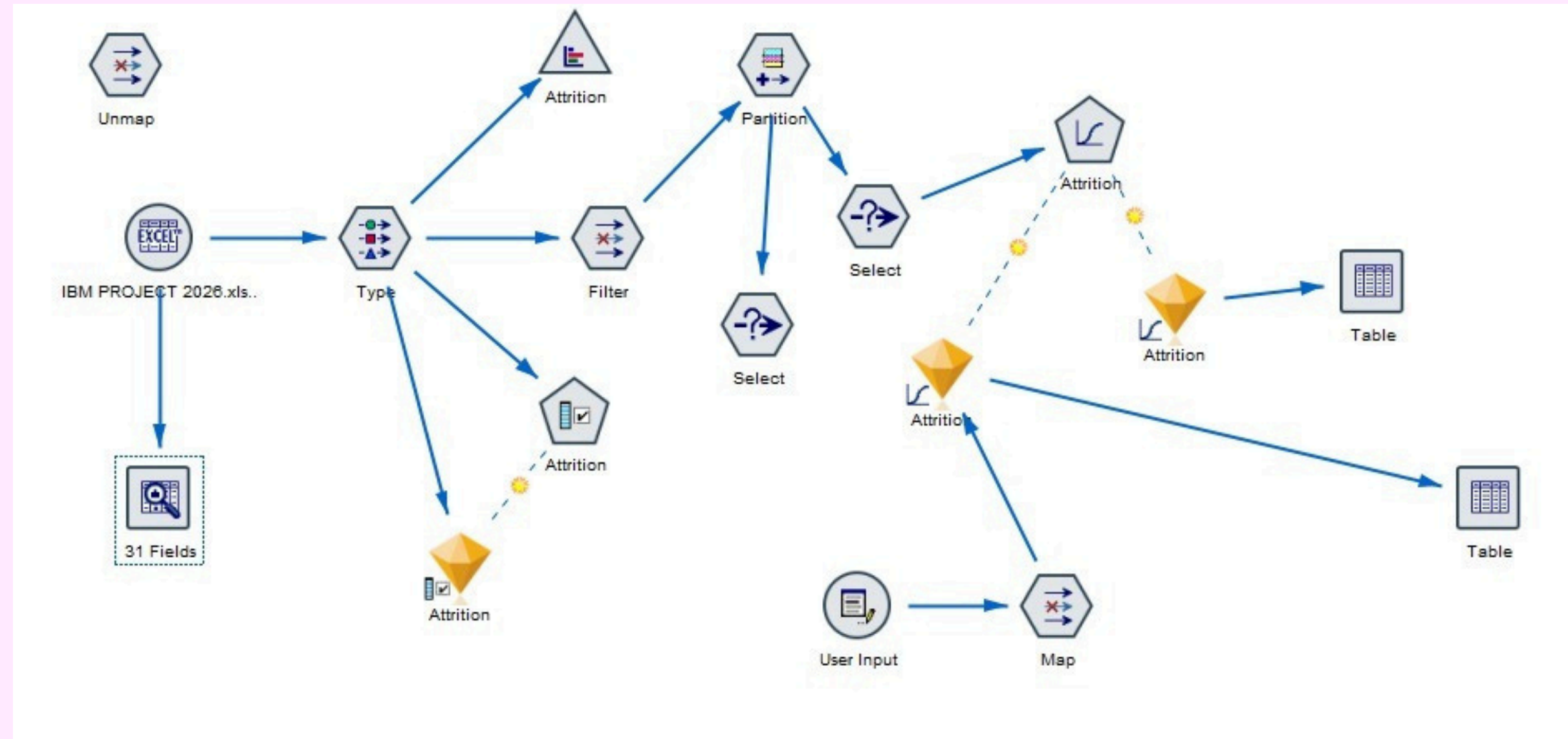
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Opportunities

Enable data-driven HR decisions to reduce turnover, improve employee satisfaction, and enhance workforce planning.

Technologies Used

IBM SPSS Modeler powers our data mining and predictive modeling workflow, utilizing ML classification models for attrition prediction. Key SPSS modules include Type Node, Filter Node, Modeling Node, and Evaluation Node. Our CSV dataset features: Age, Department, Job Role, Monthly Income, Work Experience, Overtime, Job Satisfaction, and Attrition Status.

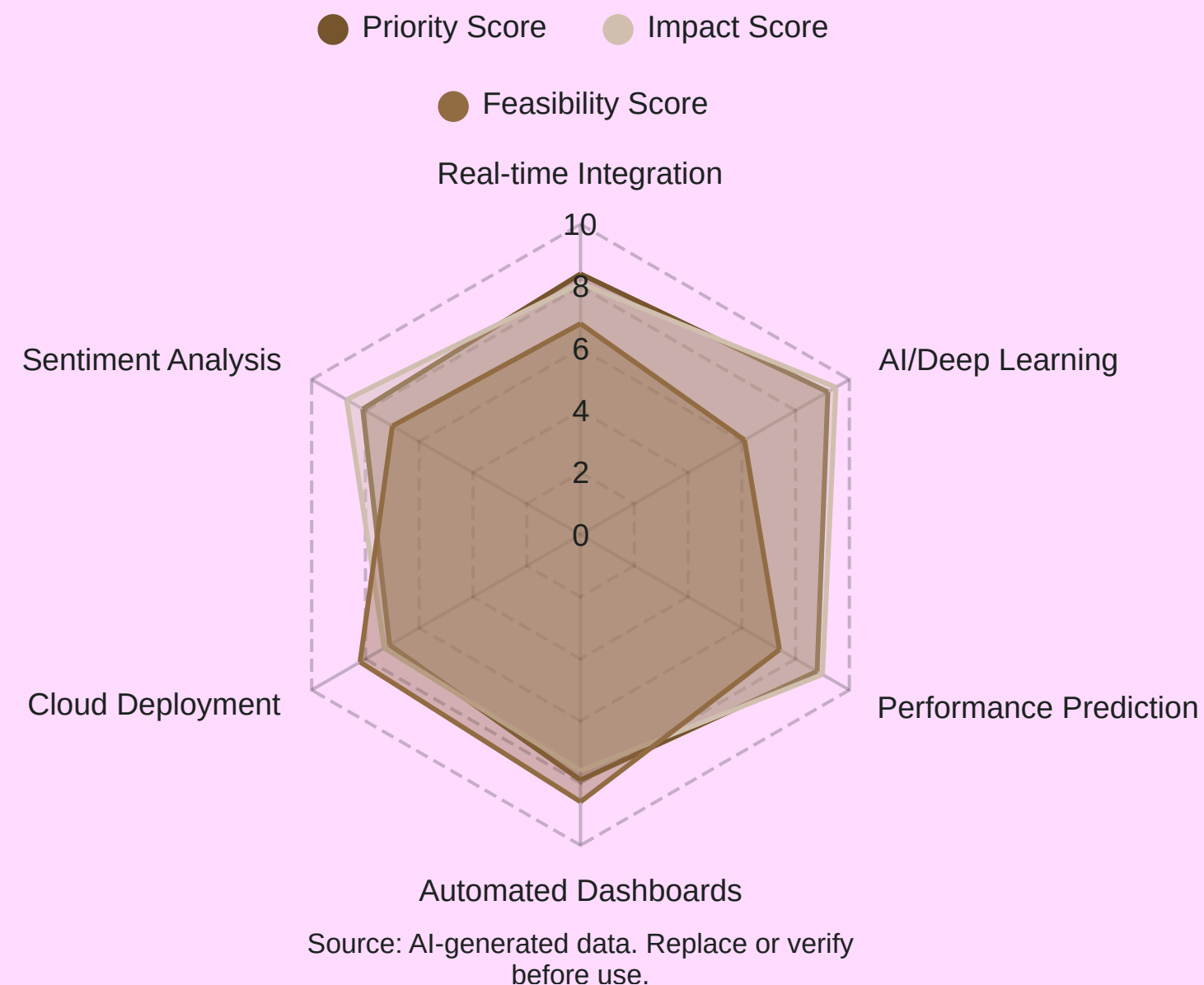


Future Scope

Future Enhancements

Innovation

- Real-time HR system integration
- Advanced AI & Deep Learning models
- Employee performance prediction
- Automated HR dashboards
- Cloud deployment for large orgs
- Sentiment analysis from feedback



Possible Improvements

Growth

- Higher accuracy with larger datasets
- Attendance & payroll integration
- Mobile-friendly HR analytics app
- Enhanced data visualization
- Cross-platform compatibility
- Scalable architecture



Conclusion

This predictive model empowers organizations to understand employee behavior and anticipate attrition effectively. IBM SPSS Modeler enables smarter HR decisions, improved employee satisfaction, and reduced turnover rates. Predictive analytics is the key to proactive retention strategies in modern HR management.

Thank You!